

Cardiac Patient Monitoring System — Sprint 3

Sprint 3 Goal

Improve the performance of the Cardiac Patient Monitoring System API by measuring database query behavior on important list endpoints, identifying and reducing N+1 query problems, and verifying measurable improvements in database query count.

Sprint 2 Retrospective Action

Write an explicit ownership-check test for every new patient-specific resource endpoint.

Performance Targets

Sprint 3 focuses on measuring and improving database query performance in important API list endpoints.

The initial performance targets are:

- Enable EF Core query logging in the development environment.
- Measure the actual number of SQL queries generated by important list endpoints.
- Test endpoints using realistic data volumes of 50+ records.
- Identify at least one genuine N+1 query problem.
- Reduce the query count of the identified N+1 endpoint to a small fixed number of queries.
- Compare query counts before and after optimization.

N+1 Diagnosis Plan

Sprint 3 will begin by measuring the actual database query behavior of important list endpoints.

The diagnosis process will include:

- Enable EF Core query logging in the development environment.
- Seed the database with at least 50 records in the primary tables.
- Run 2-3 important list endpoints.
- Count the actual SQL queries generated by each endpoint.
- Identify repeated related-data queries that indicate an N+1 problem.
- Document at least one confirmed N+1 issue as a Sprint 3 backlog task.

Sprint Backlog

Aa Name	⚙ Status
<u>Enable EF Core query logging in the development environment</u>	Done
<u>Seed the database with 50+ realistic test records</u>	Done
<u>Measure query counts for 2-3 important list endpoints</u>	Done
<u>Identify at least one genuine N+1 problem</u>	Done

Aa Name	⚙ Status
<u>Document the confirmed N+1 issue as a backlog task</u>	Done
<u>Optimize VitalSigns diagnostic N+1 query from 51 queries to a small fixed number</u>	To Do
<u>Optimize the identified N+1 query.</u>	To Do
<u>Compare query counts before and after optimization</u>	To Do
<u>Carry forward the Sprint 2 ownership-check testing improvement action</u>	To Do